STACKS, MODULI OF CURVES: LEMMA-SMOOTH-CURVES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1737-1760 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 1762-1787, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 17 of queue_moduli-curves.json (pb=1140, stmt 1737-1760, proof 1762-1787).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00SB: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1787–1787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00T5: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1787–1787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E1F: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1008–1019. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZI: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1787–1787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *There exist an open substacks*

$$\mathcal{C}urves^{smooth,1} \subset \mathcal{C}urves^{smooth} \subset \mathcal{C}urves$$

such that

(1) *given a family of curves $f : X \to S$ the following are equivalent*
   (a) *the classifying morphism $S \to \mathcal{C}urves$ factors through $\mathcal{C}urves^{smooth}$, resp. $\mathcal{C}urves^{smooth,1}$,*
   (b) *f is smooth, resp. smooth of relative dimension 1,*

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(2) *given X a scheme proper over a field k with $\dim(X) \leq 1$ the following are equivalent*
   (a) *the classifying morphism $\mathrm{Spec}(k) \to \mathcal{C}urves$ factors through $\mathcal{C}urves^{smooth}$, resp. $\mathcal{C}urves^{smooth,1}$,*
   (b) *X is smooth over k, resp. X is smooth over k and X is equidimensional of dimension 1.*

Proof. To prove the statements regarding $\mathcal{C}urves^{smooth}$ it suffices to show that given a family of curves $f : X \to S$, there is an open subscheme $S' \subset S$ such that $S' \times_S X \to S'$ is smooth and such that the formation of this open commutes with base change. We know that there is a maximal open $U \subset X$ such that $U \to S$ is smooth and that formation of U commutes with arbitrary base change, see Morphisms of Spaces, Lemma 0DZI. If $T = |X| \setminus |U|$ then $f(T)$ is closed in S as f is proper. Setting $S' = S \setminus f(T)$ we obtain the desired open.

Let $f : X \to S$ be a family of curves with f smooth. Then the fibres X_s are smooth over $\kappa(s)$ and hence Cohen-Macaulay (for example you can see this using Algebra, Lemmas 00T5 and 00SB). Thus we see that we may set

$$\mathcal{C}urves^{smooth,1} = \mathcal{C}urves^{smooth} \cap \mathcal{C}urves^{CM,1}$$

and the desired equivalences follow from what we've already shown for $\mathcal{C}urves^{smooth}$ and Lemma 0E1F. □